

### Yesterday's Solution, Today's Structure: Why Dialysis Stays In-Center

**The law, and the claim.** Of the eleven laws in Chapter 4, "Today's problems come from yesterday's solutions" best accounts for healthcare's most stubborn failures. I apply it to a case I can specify precisely: the dominance of in-center hemodialysis in the United States, where roughly one dialysis patient in eight is treated at home. The in-center default is not a preference anyone holds but a structure an earlier and entirely defensible solution built, and the structure now sustains itself. In Senge's vocabulary this is a **Shifting the Burden** archetype whose outward behavior reads as Fixes That Fail.

**The causal structure.** In 1972 Congress extended Medicare to end-stage renal disease. Before it, people who could not pay for dialysis died; after it, they did not. The mechanism chosen was payment for a treatment delivered in a facility, and it is that mechanism, not the entitlement, that propagated:

**B1 — symptomatic:** incident patient → in-center chair → kidney failure treated (days)

**B2 — fundamental:** incident patient → home training → home therapy → same result, plus preserved residual function and lower cost (weeks to months)

**R1 — dependence:** in-center volume → clinic capital, staffing ratios, referral habit, fellowship exposure → in-center is the default and the fastest route → in-center volume

**Atrophy arm:** R1 growth → home training capacity declines → B2 slower and riskier → B2 chosen less → R1 grows

B1 and B2 relieve the same symptom. B1 is faster, so under pressure it wins, and every win feeds R1. What makes this Shifting the Burden rather than ordinary competition is the atrophy arm: R1 does not merely crowd B2 out, it degrades B2's capability, so the fundamental solution becomes genuinely harder to choose over time rather than merely less popular.

**Delays, which are why nobody sees it.** Three delays run at different scales. A placement decision resolves in days. Building home program capability takes months to years, since a home training nurse is not interchangeable with a floor nurse. The structural delay runs in decades: clinic leases, corporate footprints, fellowship curricula. Because the reinforcing loop's delay is by far the longest, it was invisible to Congress in 1972 and is invisible to the nephrologist making a placement this afternoon. Cause and effect are separated here by roughly fifty years.

**Short term against long term.** In the short term the symptomatic path wins on every metric anyone is held to: time to first treatment, nursing hours per patient, chair utilization, revenue per patient-month. In the long term it yields higher total cost of care, faster loss of residual kidney function, transport burden carried by patients, and a workforce that can no longer deliver the alternative at scale. No short-term metric registers that last effect, which is why the trade is made repeatedly and never deliberately.

**Policy resistance.** Two real interventions show the structure pushing back. Requiring that patients be educated on all modalities raised awareness without touching the capability that binds, and home share barely moved. Payment models rewarding home starts produced home starts — and a meaningful share of those patients returned to in-center within the year, so the gain landed in the wrong measure. Each fix acted on an arm the structure could route around.

**Testable measures.** A real change would be distinguishable from a metric artifact by:

- home modality share at 90 and 365 days after incidence, never at start;
- home-to-center transfer rate within 12 months, risk-adjusted;
- home training nurse FTE per 100 home patients, and days a funded post sits vacant;
- median days from modality decision to first home treatment;
- residual kidney function at 12 months, stratified by modality;
- total cost of care per patient-year, transport included.

**The high-leverage intervention, and where it acts.** Pay for home training *capacity* rather than for training events: a per-FTE payment for credentialed home training nurses, accounted separately from treatment revenue and tied to twelve-month home retention rather than to home starts. It acts on the atrophy arm — the link that makes B2 slow and risky — not on the symptomatic loop, which is why it requires no clinician to decide differently. It is also the least obvious lever available; the obvious ones, patient education and per-start payment, both act on inputs to B1.

**Anticipated second-order effects.** Favorably, shortening B2's delay makes home the fast path as well as the cheap one, inverting the pressure that now favors B1, and trainees rotating through functioning home programs reverse the skill atrophy a fellowship generation later. Unfavorably, a per-FTE payment invites staffing to the payment rather than to demand, and retention-linked payment creates an incentive to select lower-risk patients into home therapy — improving the measure while worsening equity. The second is the serious one, mitigated rather than solved by risk-adjusting retention and publishing case mix.

**Assumptions, stated so they can be attacked.** That the gap is supply-side, meaning home therapy is suited to materially more patients than receive it; if preference or housing is the binding constraint, this intervention will not move the outcome. That training capacity rather than nephrologist willingness is what binds — suggestive, not settled. That payers can observe twelve-month retention reliably enough to pay against it. And that in-center capital resists for at least a decade, so an intervention judged on a three-year horizon will be called a failure before the structure has moved.

Senge's law does not say the 1972 solution was wrong. It says solutions propagate structure, and structure outlives the problem it was built for. Chapter 4's discipline is to name the structure before adding one more fix to it.